JOURNAL OF NEUROTRAUMA 33:766–776 (April 15, 2016)
© Mary Ann Liebert, Inc.
DOI: 10.1089/neu.2015.4070

Recovery from Mild Traumatic Brain Injury in Previously Healthy Adults

Heidi Losoi,[1,2] Noah D. Silverberg,[3] Minna Wäljas,[1] Senni Turunen,[1] Eija Rosti-Otajärvi,[1] Mika Helminen,[4] Teemu M. Luoto,[1] Juhani Julkunen,[2] Juha Öhman,[1] and Grant L. Iverson[5]

Abstract

This prospective longitudinal study reports recovery from mild traumatic brain injury (MTBI) across multiple domains in a carefully selected consecutive sample of 74 previously healthy adults. The patients with MTBI and 40 orthopedic controls (i.e., ankle injuries) completed assessments at 1, 6, and 12 months after injury. Outcome measures included cognition, post-concussion symptoms, depression, traumatic stress, quality of life, satisfaction with life, resilience, and return to work. Patients with MTBI reported more post-concussion symptoms and fatigue than the controls at the beginning of recovery, but by 6 months after injury, did not differ as a group from nonhead injury trauma controls on cognition, fatigue, or mental health, and by 12 months, their level of post-concussion symptoms and quality of life was similar to that of controls. Almost all (96%) patients with MTBI returned to work/normal activities (RTW) within the follow-up of 1 year. A subgroup of those with MTBIs and controls reported mild post-concussion-like symptoms at 1 year. A large percentage of the subgroup who had persistent symptoms had a modifiable psychological risk factor at 1 month (i.e., depression, traumatic stress, and/or low resilience), and at 6 months, they had greater post-concussion symptoms, fatigue, insomnia, traumatic stress, and depression, and worse quality of life. All of the control subjects who had mild post-concussion-like symptoms at 12 months also had a mental health problem (i.e., depression, traumatic stress, or both). This illustrates the importance of providing evidence-supported treatment and rehabilitation services early in the recovery period.

Key words: cognitive; head injury; outcome; post-concussion syndrome; traumatic brain injury

Introduction

There is considerable heterogeneity across studies of clinical outcome from mild traumatic brain injury (MTBI),[1] with conclusions ranging from full recovery in most patients to high rates of chronic problems. Methodological inconsistencies have likely contributed to these discrepant findings. According to a recent synthesis by the International Collaboration on MTBI prognosis,[2] there still exists many methodological concerns and gaps in the knowledge, such as: (1) inconsistent definitions of MTBI; (2) inadequate control for confounding factors; (3) the lack of a nonhead injury comparison group; (4) overlooking selection and attrition biases; (5) poor description of the source populations and samples; (6) use of cross-sectional designs or prospective studies with insufficiently long follow-up (less than 12 months post-injury); and (7) outcome measures with inadequate or unknown reliability, validity, and responsiveness.[2] Well-designed confirmatory studies that address these methodological limitations are needed to enhance understanding about consequences of MTBI.[3]

Another major barrier to understanding clinical outcome from MTBI is that ''outcome'' is not universally defined. Different dimensions of outcome from MTBI have been investigated, including post-concussion symptoms and neuropsychological, psychiatric, and functional outcomes. To date, post-concussion symptom recovery has been the most studied. A systematic review based on six studies evaluating outcome from MTBI up to 6 months concluded that post-concussion symptoms (e.g., headache, fatigue, concentration difficulty) persist beyond 6 months in 14–26% of the patients.[1] The authors of the review conclude, however, that these results need confirmation, because none of these studies included control groups, for example.[1] Controlled studies have produced mixed findings as to whether long-term post-concussion symptom reporting is similar or higher in patients with MTBI versus nonhead injury.[4]

[1]Department of Neurosciences and Rehabilitation, Tampere University Hospital, Tampere, Finland.
[2]Institute of Behavioural Sciences, University of Helsinki, Helsinki, Finland.
[3]Division of Physical Medicine and Rehabilitation, GF Strong Rehab Centre, University of British Columbia, Vancouver, British Columbia, Canada.
[4]School of Health Sciences, University of Tampere and Science Center, Pirkanmaa Hospital District, Tampere, Finland.
[5]Department of Physical Medicine and Rehabilitation, Harvard Medical School, Charlestown Navy Yard, Charlestown, Massachusetts; Spaulding Rehabilitation Hospital; and Red Sox Foundation and Massachusetts General Hospital Home Base Program, Boston, Massachusetts.

Neuropsychological outcomes from MTBI have also been extensively studied. Multiple systematic reviews have consistently reported neuropsychological recovery by 90 days.[5] Despite this positive average prognosis, a subgroup of patients with MTBI might remain chronically impaired. The size and existence of this subgroup still remains debatable.[5–7] A relatively small number of studies have examined functional outcomes, primarily return to work (RTW). The best available evidence suggests that 80–95% of patients RTW within 3 to 6 months after MTBI and that MTBI is not considered a significant risk factor for delayed RTW.[8]

A few studies on psychiatric outcomes indicate that MTBI is associated with increased risk for psychiatric illness and suicide.[3] Post-traumatic stress disorder (PTSD) appears particularly common after MTBI, with prevalence in civilian patients estimated at 12–30%.[9] Post-injury depression prevalence rates of 7%[10] and 11%[11] have been reported in patients with MTBI. It is unclear if the risk for these disorders is higher after MTBI in comparison with a nonhead injury.[10,11] Factors that mediate or moderate the risk between MTBI and psychiatric outcomes are not fully understood.

Most previous studies document recovery in only one or a limited range of outcome domains. The timing of recovery across post-concussion symptom, neuropsychological, psychiatric, and functional domains and the relationship between these domains is poorly understood. As well, other important outcomes have been neglected. Quality of life has been considered a useful frame of reference to evaluate outcome after MTBI.[12] There is limited information concerning the quality of life of patients with MTBI[13] and especially the longitudinal course of quality of life after MTBI.[12]

The aim of this study is to thoroughly and prospectively characterize recovery from MTBI in a sample of previously healthy adults by addressing many of the limitations in the current MTBI literature. We carefully selected a homogenous sample of patients with MTBI and excluded those with possible confounding factors, such as previous brain injuries, mental health or substance abuse problems, or other general health or neurological problems that could have an adverse effect on cognition or psychological health.

We used a prospective research design involving assessing the patients at 1, 6, and 12 months after injury. We also included a nonhead injury trauma control group and assessed outcome broadly by measuring not only the MTBI symptoms, but also the cognitive, psychiatric, quality of life, and RTW outcome. Studying clinical outcome comprehensively provides a holistic view of recovery from MTBI and allows preliminary examination of the relationships between outcome domains.

Methods

Participants

This study is a part of the Tampere Traumatic Head and Brain Injury Study, which broadly aimed to identify factors affecting long-term outcome from MTBI. Ethics approval for the study was granted from the Ethics Committee of Pirkanmaa Hospital District, Finland, and each participating subject had signed written informed consent according to the Declaration of Helsinki.

Participants were prospectively enrolled from consecutive cases at the emergency department (ED) of the Tampere University Hospital between August 2010 and July 2012. All consecutive patients with head computed tomography (CT) because of acute head injury ($n=3023$) served as the initial cohort, and they were screened to obtain a sample of working aged adults without pre-injury medical, psychiatric, or neurological problems who sustained a MTBI, and who probably could be reached for an outcome visit. The indications for acute head CT were based on the Scandinavian guidelines for initial management of minimal, mild, and moderate head injuries.[14]

Subjects were included in the study if they: (1) met MTBI criteria of the World Health Organization's Collaborating Centre for Neurotrauma Task Force,[15] (2) were aged between 18 and 60 years, and (3) were residents of the hospital district. Subjects were excluded if they had: (1) pre-morbid neurological problems ($n=771$, 25.5%), (2) previous psychiatric problems ($n=916$, 30.6%), (3) past TBI ($n=311$, 10.5%), (4) regular psychoactive medication use ($n=835$, 27.6%), (5) neurosurgery ($n=168$, 5.6%), (6) problems with vision or hearing ($n=47$, 1.6%), (7) first language was not Finnish ($n=74$, 2.4%), (8) the time interval between injury and arrival to the ED was more than 72 h ($n=506$, 16.7%), and/or (9) they declined to participate in the study ($n=105$, 3.5%).

There was major overlap in the causes of exclusion; some patients had multiple reasons (e.g., age, MTBI criteria not met, and a neurological or psychiatric problem). Patient enrollment details are discussed thoroughly in our previous publication.[16]

Control subjects were orthopedically injured patients evaluated in the same ED as the MTBI sample. Consecutive patients ($n=609$) with ankle injury (bone fracture or distension) were screened for inclusion. The same study criteria used with the MTBI sample were applied to the controls, as applicable. Control subjects were enrolled in an age and sex stratified manner, with five men and five women in the following age groups: (1) 18–30 years, (2) 31–40 years, (3) 41–50 years, and (4) 51–60 years. Age stratification was performed as part of a separate larger study to produce normative age group data for a magnetic resonance imaging (MRI) study (i.e., for diffusion tensor imaging [DTI]).

By applying the above-mentioned inclusion and exclusion criteria, 75 patients with MTBI and 40 trauma controls were recruited. One patient with MTBI did not participate in the follow-up and was excluded from this study. Therefore, the final MTBI cohort of this study included 74 patients.

Procedure

Acute clinical assessment and neuroimaging.

A clinical assessment of the participants was performed by a research physician (TML) in the ED or, if the patient was discharged before that, in a separate hospital visit as soon as possible (mean time in hours between the injury and acute clinical assessment = 48.1, range = 2.0–241.0, interquartile range [IQR] = 19.4–66.4). The assessment included a thorough interview of past health, diagnosed medical conditions, medication use, head injury history, alcohol consumption, and drug and narcotics abuse history. None of the patients in this study had regular psychoactive medication use.

Injury-related data consisted of time of injury, mechanism of injury, and alcohol intoxication at the time of injury. The presence and duration of possible loss of consciousness was recorded using information given by eyewitnesses and ambulance personnel where available. The presence and duration of retrograde-and post-traumatic amnesia were evaluated using the Rivermead PTA Protocol and the Galveston Orientation and Amnesia Test (GOAT).[17] Glasgow Coma Scale (GCS) scores were collected from the ambulance (if available) and the ED records.

A complete neurological examination, including the cranial and spinal nerves, coordination, pronator drift, balance, and diadochokinesis was completed. The overall severity of physical injuries was assessed using the Injury Severity Score (ISS).[18] All patients with MTBI and controls underwent 3T MRI of the brain (Siemens Trio, Siemens AG Medical Solutions, Erlangen, Germany). The MRI protocol included sagittal T1-weighted three-dimensional (3D) inversion recovery (IR) prepared gradient echo, axial T2 turbo spin echo, conventional axial and high resolution sagittal FLAIR (fluid-attenuated inversion recovery), axial T2*, axial SWI (susceptibility weighted imaging), DWI (diffusion weighted imaging) and DTI series. Mean time between injury and MRI was 5.8 days (standard deviation [SD] = 2.5, Md = 5.4).

Follow-up. Participants filled out self-report questionnaires about demographic variables, post-concussion symptoms, fatigue, insomnia, pain, post-traumatic stress, and quality of life on an internet-based platform at 1, 6, and 12 months post-injury. The patients unable to complete the questionnaires online completed them at the follow-up visit ($n = 9$ at 1 month after injury and $n = 7$ at 6 months after injury) or by mail at 12 months after injury ($n = 4$). Depressive symptoms were evaluated at a neuropsychological follow-up visit only at 1 and 6 months post-injury. Neuropsychological examination was conducted for the patients with MTBI and for the controls at 1 month after injury and for the MTBI group at 6 months.

Outcome measures

The Extended Glasgow Outcome Scale (GOS-E).[19] This is a widely used scale based on a structured interview. It reports the patient's recovery on a scale from 1–8 with scores 1–4 reflecting severe disability, scores 5–6 moderate disability, and 7–8 good recovery.

The Rivermead Post-Concussion Symptom Questionnaire (RPCSQ).[20] This was used to evaluate post-concussion symptoms. It is a 16-item self-report questionnaire measuring the severity of common post-concussion symptoms on a 5-point Likert scale. The patients rate the presence of the symptoms on a scale from 0–4. A total score is a sum of the items. The presence of post-concussion syndrome was defined using *International Statistical Classification of Diseases and Related Health Problems 10th Revision* (ICD-10) diagnostic criteria for post-concussion syndrome (PCS).[21]

Participants were determined to have met the criteria for PCS if they reported symptoms on the RPCSQ in at least three of the following ICD-10 symptoms categories: (1) headaches, dizziness, general malaise, excessive fatigue, or noise intolerance; (2) irritability, emotional lability, depression, or anxiety; (3) subjective complaints of concentration or memory difficulty; (4) insomnia; (5) reduced tolerance to alcohol; (6) preoccupation with these symptoms and fear of permanent brain damage.

The first four of these symptom categories could be assessed by using the RPCSQ. Two thresholds for symptom reporting were considered: mild or greater severity (2–4 points) and moderate or greater severity (3–4 points), and these are referred to as mild PCS or moderate PCS, respectively.

The Barrow Neurological Institute Fatigue Scale (BNI-FS). This was used to assess fatigue. It is a reliable and valid[22] 11-item self-report questionnaire designed to assess fatigue after TBI.[23] The total BNI-FS score is the sum of the first 10 items (min = 0, max = 70).

The Insomnia Severity Index (ISI). This is a reliable and valid seven-item self-report questionnaire assessing the patient's perception of his or her insomnia.[24] The total score ranges from 0 to 28.

The Pain Subscale of the Ruff Neurobehavioral Inventory (RNBI).[25]. This was used to evaluate pain. The subscale comprises six items rated on a 4-point scale (1–4), and the total score ranges from 6–24.

PTSD-Checklist-Civilian Version (PCL-C).[26] Symptoms of PTSD were assessed in civilians with this reliable and valid scale.[27] The possible scores of the scale range from 17–85. The participant was defined as having PTSD if the total score of PCL-C was greater than 50 or the criteria for PTSD in *Diagnostic and Statistical Manual of Mental Disorders, 4th Edition* (DSM-IV)[28] was fulfilled, and possible PTSD if the total score of PCL-C was greater than 35.

Beck Depression Inventory- Second Edition (BDI-II).[29] This self-report questionnaire in which the total score ranges from 0 to 63 was used to assess depressive symptoms at 1 and 6 months post-injury. Depression was diagnosed using selected items of the questionnaire. Ten of the 21 symptoms from the BDI-II, believed to have the least overlap with symptoms of MTBI and being most representative of depression, were selected. These symptoms were: sadness, loss of interest, loss of pleasure, pessimism, past failure, guilt feelings, punishment feelings, self-criticalness, crying, and suicidal thoughts or wishes. The algorithm for defining depression was as follows: (1) sadness or loss of pleasure must be endorsed, and 3 or more of the 10 core symptoms must be endorsed, and the BDI total score had to be 10 or greater; or (2) 5 or more of the 10 selected symptoms had to be endorsed. Patients meeting either criterion were included in the depression group, and the remaining subjects were classified as not depressed.

The Resilience Scale (RS).[30] This 25-item self-report questionnaire is designed to assess the degree of an individual's resilience. The possible total scores range from 25–175 with higher scores reflecting higher resilience. Total score ≤130 is considered relatively low. The RS has been reported to be a reliable and valid tool to measure resilience.[30]

Quality of life. According to the recommendations by the TBI Consensus Group,[31] the assessment of quality of life after TBI should include both a disease-specific and a generic instrument. The disease-specific quality of life was measured in this study by the Quality of Life after Brain Injury (QOLIBRI), which is a reliable[32] and valid[33] health-related quality-of-life instrument specifically developed for TBI. It contains 37 items rated on a 5-point Likert scale; the possible total score thus ranges from 37–185. In the present study, controls were asked to rate their quality of life with this questionnaire while considering their orthopedic injury. The Satisfaction with Life Scale (SWLS) was used to measure generic and global life satisfaction. The scale has favorable psychometric properties.[34] It contains five items on a 7-point Likert scale, and the possible total score ranges from 5–35.

Neuropsychological measures. The Rey Auditory Verbal Learning Test (RAVLT; total number of words recalled in trials 1–5, recall after interference, and recognition after 30 min)[35] was used to assess verbal memory. The Stroop Test (Golden version, number of items completed in color-word interference trial),[35] Trail Making Test Parts A and B (TMT; time in seconds),[36] phonemic (P/A/S) and semantic (animals) verbal fluency (number of words in 1 min)[37] were used to assess executive functions. Motor speed was assessed with the Finger Tapping Test (mean number of taps in five consecutive 10 sec trials for the dominant and non-dominant hand).[38]

The following subtests from the Wechsler Adult Intelligence Scale–Third Edition (WAIS-III)[39] were used: Information to assess verbal intelligence, Digit Span to assess attention and working memory, and Digit-Symbol Coding and Symbol Search to assess processing speed. Cognitive impairment was defined using published Finnish (WAIS-III)[39] and international[40] normative data. The participant was defined as having cognitive impairment if 4 or more of the 14 primary test scores on the neuropsychological measures were at least one standard deviation below average. This criterion is based on past studies examining the base rates of low scores in healthy adults when multiple scores are considered simultaneously.[41–43]

It is common for healthy adults to obtain some low scores when multiple tests are administered. Adults with above average intelligence obtain fewer low scores than adults with average or below average intelligence. As noted by Iverson and colleagues,[44] it is common for adults of average intelligence to have 20–30% of their test scores ≤1 SD from the mean, and it is common for adults with above average intelligence to have approximately 15% of their test scores in this range.

RTW. This is defined as the duration of sick leave in days, which for the students and unemployed means the time before returning to their normal activities. Information about RTW was obtained from interviews and medical files. The primary reason for the sick leave was assessed collaboratively by the study physician and neuropsychologist, based on the symptoms and findings, to be the MTBI, bodily injuries, or both. Data about RTW is only available for patients with MTBI.

Data analyses

The statistical analyses were conducted using the IBM SPSS Statistics for Windows, Version 22.0. The proportions were compared with the Fisher exact test, and continuous variables were tested with Mann-Whitney or *t* tests. Cohen d was used as an effect size measure for the standardized difference between the MTBI and control group.

Results

Demographic and descriptive characteristics

The majority ($n = 58$, 78.4%) of the patients with MTBI were working at the time of the injury. The rest were full-time students ($n = 13$, 17.6%) or unemployed ($n = 3$, 4.0%). The mechanisms of injury for the MTBI group ($n = 74$) were as follows: sports ($n = 13$, 17.6%), falls from a height ($n = 12$, 16.2%), car accidents ($n = 12$, 16.2%), bicycle accidents ($n = 11$, 14.9%), ground-level falls ($n = 10$, 13.5%), motorcycle accidents ($n = 5$, 6.8%), violence-related injuries ($n = 4$, 5.4%), and other ($n = 7$, 9.5%). All the patients with MTBI scored 80 points or greater on the GOAT, and their GCS was 14–15.

The characteristics of the patients with MTBI and the controls are described in Table 1. There were no statistically significant differences between the MTBI group and controls in age, education (in years), or sex. None of the controls had loss of consciousness or traumatic findings on MRI. The MTBI group had slightly more physical injuries (ISS) than the controls, but the difference did not reach statistical significance.

There were no significant differences between the MTBI group and the controls in the time from injury to completing the questionnaires (at 1 and 12 months) or attending to the neuropsychological examination at 1 month after injury. The difference in days to completing the 6-month questionnaires was statistically significant but subtle (7.7 days). It can be considered clinically irrelevant.

There was some loss of participants during the 12-month follow-up ($n = 14$, 18.9% in patients with MTBI and $n = 11$, 25.6% in controls). The possible effects of attrition were examined using demographic (age, sex, and education) and trauma-related (GCS, and presence of loss of consciousness [LOC]) variables, self-reported symptoms (RPCSQ, BNI, ISI, BDI-II, PCL-C, QOLIBRI, SWLS, Pain Subscale of RNBI) at 1 month, and RTW. No systematic bias was found because of attrition. None of the dropouts, however, had traumatic lesions on MRI (vs. $n = 15$ of nondropouts, $p = 0.059$).

Recovery measured by the GOS-E

At 1 month after injury, GOS-E ranged from 4–8 and at 6 months from 5–8. At 1 month 67.7% and at 6 months 88.9% of the patients with MTBI were classified as having a good recovery.

Table 1. Characteristics of Patients with Mild Traumatic Brain Injury and Orthopedic Controls (Ankle Injury)

Descriptive variables	*MTBI* (n = 74)	*Controls* (n = 40)	p *value*	*Cohen d*
Sex, female: n (%)	29 (39.2)	20 (50.0)	0.266	NA
Age (years): mean (SD)	37.0 (11.8)	40.1 (12.2)	0.186	–0.26
Education (years): mean (SD)	14.2 (3.1)	14.1 (2.8)	0.891	0.03
Injury Severity Score: mean (SD)	3.9 (3.2)	2.6 (1.5)	0.056	0.48
Injury Severity Score without head injuries: mean (SD)	2.2 (2.9)	2.6 (1.5)	0.013	0.14
Traumatic findings on CT: n (%)	7 (9.5)	NA	NA	NA
Traumatic Findings on MRI: n (%)	15 (20.3)	0	0.001	NA
Diffuse axonal injury	7 (9.5)			
Diffuse axonal injury and subdural hemorrhage	1 (1.4)			
Subdural hemorrhage	1 (1.4)			
Subdural effusion	1 (1.4)			
Subarachnoid hemorrhage	1 (1.4)			
Contusion and subdural hemorrhage	2 (2.7)			
Contusion	2 (2.7)			
Loss of consciousness (LOC): n (%)	27 (36.5)	0	0.000	NA
Duration of LOC (min): mean (SD)	0.9 (2.2)	NA	NA	NA
Post-traumatic amnesia (PTA): n (%)	68 (92.0)	NA	NA	NA
Duration of PTA (h): mean (SD)	2.6 (3.4)	NA	NA	NA
Neurological deficit (e.g., the cranial and spinal nerves)	17 (22.7)	NA	NA	NA
Days to 1 month questionnaire: mean (SD)	25.2 (4.2)	23.9 (3.4)[1]	0.091	0.33
Days to 1 month cognitive testing: mean (SD)	29.0 (4.9)	30.6 (9.3)	0.832	–0.24
Days to 6 month questionnaires: mean (SD)	185.5 (11.3)[2]	193.2 (18.4)[2]	0.004	–0.55
Days to 6 month cognitive testing: mean (SD)	193.5 (47.0)[3]	NA	NA	NA
Days to 12 month questionnaires: mean (SD)	380.0 (16.0)[4]	393.1 (67.6)[5]	0.854	–0.32

MTBI, mild traumatic brain injury; SD, standard deviation; NA, not available.
[1]1 case missing, [2]3 cases missing, [3]9 cases missing, [4]14 cases missing, [5]11 cases missing.
The proportions were compared with the Fisher exact test, and continuous variables were tested with Mann-Whitney or *t* tests.

Table 2. Outcome from Mild Traumatic Brain Injury and Ankle Injury

	1 month			*6 months*			*12 months*		
	MTBI	*Controls*	p *value*	*MTBI*	*Controls*	p *value*	*MTBI*	*Controls*	p *value*
Sample size	74	40	–	69	37	–	60	29	–
Good recovery on GOS-E N (%)	50 (67.6)	NA	–	56 (88.9)[6]	NA	–	NA	NA	–
Post-Concussion Syndrome Mild N (%)	23 (31.1)	8 (20.5)[1]	0.273	17 (24.6)	5 (13.5)	0.312	16 (26.7)	5 (17.2)	0.431
Post-Concussion Syndrome Moderate N (%)	4 (5.4)	1 (2.6)[1]	0.658	3 (4.3)	0	0.549	3 (5.0)	1 (3.4)	1.000
Cognitive impairment N (%)	14 (19.4)[2]	4 (10.5)[2]	0.290	6 (9.2)[4]	NA	NA	NA	NA	–
Depression N (%)	12 (16.4)[1]	6 (15.8)[2]	1.000	5 (7.7)[4]	6 (18.8)[5]	0.167	NA	NA	–
Possible PTSD N (%)	12 (16.2)	5 (12.5)	0.784	7 (10.0)	5 (13.5)	0.536	4 (6.7)	3 (10.3)	0.675
Probable PTSD N (%)	3 (4.1)	1 (2.5)	1.000	3 (4.3)	2 (5.4)	1.000	2 (3.3)	0	1.000
Psychiatric disorder (depression or probable PTSD) N (%)	12 (16.4)[1]	6 (15.8)[2]	1.000	6 (8.0)	6 (15.0)	0.193	NA	NA	–
Return to work (RTW) %	68.0	NA	–	93.3	NA	–	96.0	NA	–

MTBI, mild traumatic brain injury; GOS-E, Glasgow Outcome Scale-Extended; NA, not available; PTSD, post-traumatic stress disorder.
The numbers represent the percentages of subjects in each group who meet criteria for the specific outcome.
[1]1 case missing, [2]2 cases missing, [3]3 cases missing, [4]4 cases missing, [5]5 cases missing, [6]6 cases missing

Table 3. Comparison of Outcome Variables of Patients with Mild Traumatic Brain Injury and Controls

	MTBI	*Controls*	p *value*	*Cohen d*
1 month	Mean (SD)	Mean (SD)		
Sample size	74	40		
Post-Concussion Symptoms (RPCSQ)	10.5 (9.6)	4.9 (6.8)[1]	0.001	0.64
Fatigue (BNI-FS)	14.6 (15.8)[1]	8.0 (7.6)[2]	0.060	0.49
Insomnia (ISI)	5.2 (5.2)[2]	4.3 (4.3)[1]	0.520	0.18
Pain Subscale (RNBI)	8.5 (2.8)	8.3 (2.5)[1]	0.654	0.07
PTSD Symptoms (PCL-C)	25.3 (8.9)	24.4 (7.4)	0.719	0.11
Depressive Symptoms (BDI-II)	5.4 (5.8)	5.1 (5.4)[1]	0.597	0.05
Resilience (RS)	139.5 (17.5)	139.5 (16.0)[1]	0.821	0.00
Quality of Life (QOLIBRI)	152.0 (27.0)	155.8 (18.3)	0.833	–0.16
Life Satisfaction (SWLS)	27.0 (6.0)	26.8 (4.8)[1]	0.964	0.04
6 months				
Sample size	71	37		
Post-Concussion Symptoms (RPCSQ)	7.3 (9.3)[2]	3.7 (6.3)	0.029	0.43
Fatigue (BNI-FS)	9.4 (13.0)	10.4 (7.8)	0.774	–0.09
Insomnia (ISI)	4.3 (4.0)	4.4 (3.8)	0.267	–0.03
Pain subscale (RNBI)	7.4 (2.3)	7.1 (2.0)	0.406	0.14
PTSD Symptoms (PCL-C)	24.0 (9.6)[1]	23.0 (8.1)	0.384	0.11
Depressive symptoms (BDI-II)	4.3 (6.2)[3]	5.0 (6.4)[4]	0.779	–0.11
Resilience (RS)	139.7 (18.8)[1]	143.3 (14.4)[1]	0.475	–0.21
Quality of Life (QOLIBRI)	157.3 (22.2)[1]	163.7 (22.5)	0.035	–0.29
Life Satisfaction (SWLS)	26.4 (5.5)	26.8 (5.2)	0.601	–0.07
12 months				
Sample size	60	29		
Post-Concussion Symptoms (RPCSQ)	6.9 (8.4)	4.4 (7.8)	0.070	0.30
Fatigue (BNI-FS)	8.1 (12.6)	8.7 (13.4)[1]	0.912	–0.05
Insomnia (ISI)	4.3 (4.1)	4.0 (4.0)	0.784	0.07
Pain Subscale (RNBI)	7.3 (2.6)	7.2 (2.3)	0.915	0.04
PTSD Symptoms (PCL-C)	23.3 (6.8)	22.7 (8.4)	0.142	0.08
Resilience (RS)	143.3 (18.4)	141.2 (19.5)[1]	0.674	0.11
Quality of Life (QOLIBRI)	158.4 (21.4)	155.7 (26.4)	0.916	0.12
Life Satisfaction (SWLS)	27.0 (5.3)	26.5 (5.7)	0.752	0.09

MTBI, mild traumatic brain injury; SD, standard deviation; RPCSQ, Rivermead Post-Concussion Symptom Questionnaire; BNI-FS, Barrow Neurological Institute Fatigue Scale; ISI, Insomnia Severity Index; RNBI; Ruff Neurobehavioral Inventory; PCL-C, PTSD-Checklist-Civilian Version; BDI-II, Beck Depression Inventory- Second Edition; RS, Resilience Scale; QOLIBRI, Quality of life after Brain Injury; SWLS, Satisfaction with Life Scale. The group comparisons were conducted with Mann-Whitney or *t* tests.
[1]1 case missing, [2]2 cases missing, [3]6 cases missing, [4]5 cases missing.

Post-concussion symptoms and PCS

At 1 and 6 months after the injury, patients with MTBI reported significantly ($p = 0.001$, Cohen $d = 0.64$; $p = 0.029$, Cohen $d = 0.43$, respectively) more post-concussion symptoms than the controls (Table 3). At 12 months ($p = 0.07$, Cohen's $d = 0.30$) the clinical significance of the difference in post-concussion-symptom reporting was small.

Based on their self-reported symptoms, 31.1% of the patients with MTBI were determined to have mild PCS at 1 month after injury, 24.6% at 6 months after injury, and 26.7% at 1 year after injury (Table 2). Mild PCS, however, was also found in a fairly large proportion of the controls as well in the follow-ups, 20.5%, 13.5%, and 17.2%, respectively. Using the reporting of at least moderate symptoms as a criterion, the prevalence of PCS at the follow-ups was much smaller: 5.4%, 4.3%, and 5.0% for the patients with MTBI and 2.6%, 0%, and 3.4% for the controls. Patients with MTBI did not meet criteria for PCS (mild or moderate) statistically significantly more often than controls at any follow-up (Table 3).

Fatigue, insomnia, and pain

There were no statistically significant differences in reporting of fatigue, insomnia, and pain at any of the follow-ups (Table 2). At 1 month after the injury, there was a trend toward significance ($p = 0.06$, Cohen $d = 0.49$) in patients with MTBI reporting more fatigue symptoms than the controls (Table 2), but at the 6- and 12-month follow-ups, there was no difference in the level of fatigue between the groups.

Cognition

At 1 month after the injury, 14 (19.4%) of the patients with MTBI and 4 (10.5%) of the controls met criteria for mild cognitive impairment (Table 3). At the 6-month follow-up, however, the rate of cognitive impairment in the MTBI group was 9.2% ($n = 6$). At the group level, at 1 month after the injury, there were no statistically significant differences in neuropsychological test performance between the patients with MTBI and controls (Table 4). At the 6-month follow-up, the memory (RAVLT total and post-interference recall, $p < 0.01$), and processing speed (Stroop, $p = 0.01$; TMT A, $p < 0.01$; WAIS-III Symbol Search, $p < 0.01$; WAIS-III Digit symbol Coding, $p < 0.01$) performance of patients with MTBI was significantly better than at 1 month after injury, likely reflecting a combination of practice effects and improvement in cognition in some people.

Mental health outcome

There were no significant differences in reporting of traumatic stress or depressive symptoms between patients with MTBI and controls (Table 2) at any of the follow-ups. At 1 month after the injury, 16.4% of the MTBI group and 15.8% of the controls, and at 6 months 8.0% of the patients with MTBI and 15.0% of the controls were defined as having a mental health problem (depression or traumatic stress) (Table 3). At 12 months, depression was not assessed and 3.3% of the patients with MTBI and zero controls met criteria for probable PTSD.

Quality of life

At 1 and 12 months after the injury, there were no statistically significant differences in quality of life (QOLIBRI) between the patients with MTBI and controls, and at 6 months after the injury, the controls reported better ($p = 0.04$) quality of life than the patients with MTBI (Table 2). In generic satisfaction with life (SWLS), there were no significant differences between the groups in any of the follow-ups.

Table 4. Comparison of Neuropsychological Variables (z-scores) of Patients with Mild Traumatic Brain Injury and Controls

	MTBI 1 months (N=74)	MTBI 6 months (N=65)	MTBI 1 month vs. MTBI 6 months		Controls (N=40)	MTBI 1 month vs. controls		MTBI 6 months vs. controls	
Neuropsychological test	M (SD)	M (SD)	p	d	M (SD)	p	d	p	d
RAVLT total recall	.18 (.98)	.74 (.93)	.00	−.60	.34 (.93)	.37	−.17	.04	.43
RAVLT post-interference recall	−.03 (.96)	.38 (.97)	.00	−.40	.20 (.85)	.22	−.25	.11	.19
RAVLT recognition	.04 (1.16)	.30 (.75)	.51	−.10	.26 (1.04)	.32	−.20	.65	.05
Stroop Color-Word	−.04 (1.48)	.17 (1.26)	.01	−.19	.29 (1.14)	.22	−.24	.84	−.10
Verbal Fluency total	−.12 (1.17)	.13 (1.31)	.02	−.17	.32 (1.25)	.07	−.37	.54	−.15
Animal Fluency total	.69 (1.47)	.80 (1.62)	.56	−.05	1.18 (1.20)	.07	−.36	.07	−.26
TMT A (seconds)	.03 (1.10)	.37 (.87)	.00	−.37	−.06 (1.04)	.66	−.03	.02	.46
TMT B (seconds)	−.31 (1.73)	−.03 (1.04)	.11	−.23	−.04 (.72)[1]	.35	−.18	.69	.01
Finger Tapping (dominant)	.24 (1.18)	.32 (1.26)	.62	−.04	.51 (1.51)	.30	−.21	.51	−.14
Finger Tapping (nondominant)	.22 (1.19)[2]	.29 (1.17)	.76	−.02	.42 (1.27)	.42	−.16	.34	−.11
WAIS-III Information	.48 (.49)[1]	NA	NA	NA	.62 (.67)	.51	−.25	NA	NA
WAIS-III Digit Span	.52 (1.54)	.75 (.87)	.43	−.12	.53 (.90)	.95	−.01	.17	.25
WAIS-III Digit-Symbol Coding	.51 (.98)	.84 (.93)	.00	−.32	.39 (1.06)[1]	.55	.12	.05	.46
WAIS-III Symbol Search	.23 (1.00)	.63 (.95)	.00	−.44	.51 (.96)	.16	−.28	.54	.13

MTBI, mild traumatic brain injury; SD, standard deviation; RAVLT, Rey Auditory Verbal Learning Test; TMT, Trail Making Test; WAIS-III = Wechsler Adult Intelligence Scale, Third edition.

[1]1 case missing, [2]2 cases missing.

The comparisons were conducted using independent samples *t* test and paired samples *t* test.

RTW

Almost all (71 of 74, 96%) of the patients with MTBI returned to work/normal activities within the follow-up period of 1 year. The median time to RTW was 16.0 days, with most persons ($n = 61$, 82.4%) returning to work within 2 months after injury (IQR = 5.0–42.0 days). The cumulative rates for RTW at each follow-up are presented in Table 2. There were 10 patients (13.3%) who did not have any sick leave after their injury. Of the 64 patients with sick leave, the primary reason for it was a brain injury for 41 (64.1%), bodily injury for 14 (18.8%), and the combination of them for 11 (17.2%) patients.

For all three patients who remained on sick leave at 12 months, the primary reason for initial sick leave was brain injury. All three were successfully followed up after the study period. One of these patients returned to work at 1 year and 2 months after the injury, one 3 years and 2.5 months after the injury, and one patient retired, not because of problems relating to MTBI, but because of problems relating to physical injuries sustained at the time of the accident.

There were 50 MTBI subjects who returned to work within 1 month, and some of these had problems identified during the 1 month follow-up. For example, 6% had mild cognitive impairment, 26% had mild PCS, 2% had moderate PCS, 14% met criteria for depression, and 2% met criteria for PTSD and 10% for possible PTSD. A large number had at least one of these problems (36%).

Recovery trajectories during a 12-month follow-up

The self-reported outcome measures are plotted over time in Figure 1. Several observations are noteworthy. The MTBI and trauma control group trajectories track closely together for most measures, except for post-concussion symptoms (RPCSQ) and

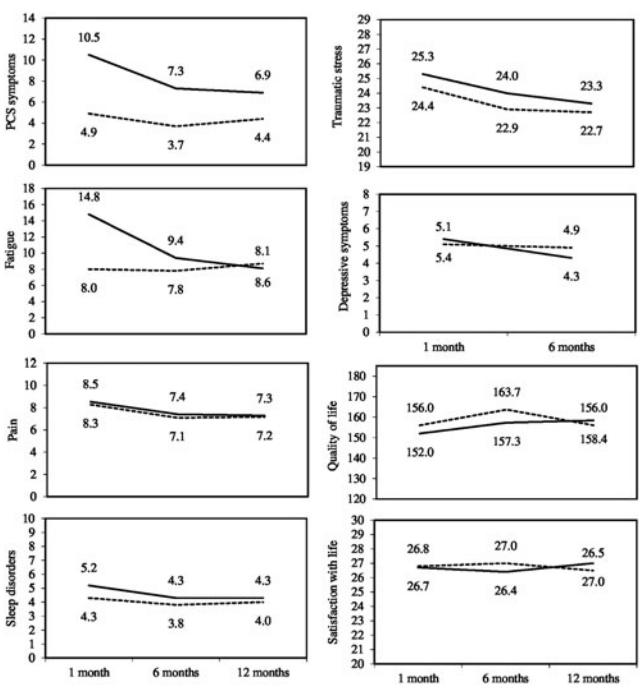

FIG. 1. Recovery trajectories of self-reported outcomes for patients with MTBI and controls.
Note: Solid lines are for those with MTBIs and dotted lines are for the control subjects. Data about depressive symptoms not available at 12 months. PCS. Post-concussion syndrome.

fatigue (BNI), which are initially more elevated in the MTBI group. The slope changes at 6 months for most outcomes, reflecting that recovery is more rapid before this point. Slight ongoing recovery beyond 6 months is evident for some measures (fatigue, post-traumatic stress), whereas others appear to plateau by 6 months (e.g., sleep, pain). Quality of life and satisfaction with life appear relatively stable over time.

The rate of symptomatic, cognitive, and vocational outcome for patients with MTBI and controls is shown in Table 3. The rate of PCS is relatively stable across follow-ups, while the proportion of those returning to work steadily increases after MTBI. Cognitive impairment in the MTBI group also appeared to steadily decrease. RTW often occurred while patients with MTBI continued to meet criteria for mild PCS.

Multidimensional recovery

PCS, cognitive impairment (at 1 and 6 months), and RTW outcomes of patients with MTBI were combined to examine the multidimensional recovery of these core domains. PCS was defined here by at least moderate symptoms, because mild symptom endorsement was commonly found in controls as well. The patients who did not have moderate PCS, did not have cognitive impairment, and had returned to work were defined as being favorably recovered. Of patients with MTBI, recovery was achieved by 62.2% at 1 month, 84.4% at 6 months, and 93.3% at 12 months (Table 3). In contrast, the patients who had moderate PCS, had cognitive impairment, and did not return to work were defined as being definitely impaired. Only one patient filled this criteria at all follow-ups (1.4–1.7%). The rest of the patients (36.4% at 1 month, 14.5% at 6 months, and 5% at 12 months) had different combinations of PCS, cognitive impairment, and RTW and can be classified as having a partial recovery.

Including mental health and mild post-concussion symptom reporting, at 1 month, 51.4% of those with MTBI (38/74) and 34.2% of control subjects (13/38) met criteria for having at least mild PCS, depression, possible PTSD, or cognitive impairment. At 6 months, 32.8% of those with MTBI (21/64) and 25.0% of controls (8/32) met criteria for having at least mild PCS, depression, or possible PTSD. At 12 months, 30.0% of those with MTBI (18/60) and 21.4% of controls (6/28) met criteria for having at least mild PCS or possible PTSD (as previously noted, the depression measure was not administered at 12 months). These differences in mental health and mild post-concussion symptom reporting between the groups were not statistically significant.

Factors associated with persistent mild PCS and delayed RTW

Further exploratory analyses were conducted with subgroups who had PCS or remained on sick leave at the 1-year follow-up. The aim of these analyses was to thoroughly characterize patients with incomplete recovery and explore possible reasons for chronic PCS and delayed RTW. Of the three patients with MTBI who had moderate PCS at 12 months, none had trauma-related abnormalities on MRI and one had cognitive impairment at 1 month post-injury (but no longer at 6 months post-injury). Of these same three patients, two reported elevated symptoms of depression and/or traumatic stress at some point.

As displayed in Table 5, patients with mild PCS at 12 months ($n = 16$, 26.7%) did not have more severe original injuries than those without PCS at 12 months ($n = 44$, 77.3%), and there were no differences in age, education, or sex between these two groups.

Patients with mild PCS at 12 months were more likely to have cognitive impairment at 1 month, but neuropsychological differences disappeared by 6 months. Patients with mild PCS at 12 months also reported greater symptoms of PCS and fatigue, and there was a trend toward them reporting greater traumatic stress at 1 month after injury.

For those who had mild PCS at 12 months, they reported significantly greater symptoms of PCS, fatigue, insomnia, traumatic stress, and depression, and worse quality of life at 6 months after injury. At 12 months, those who met criteria for mild PCS also reported greater fatigue, insomnia, and traumatic stress, lower resilience, and lower quality of life and life satisfaction than those who did not meet criteria for mild PCS.

For the 16 people with MTBIs who met criteria for mild PCS at 12 months, 10 (62.5%) met criteria for mild PCS at 1 month and 12 (75%) met criteria for mild PCS at 6 months. Six (37.5%) met criteria for possible PTSD at 1 month and three (18.8%) met this criteria at 6 months. Six (37.5%) met criteria for low resilience at 1 month. Four (25%) met criteria for depression at 1 month, but only one (6.3%) met criteria for depression at 6 months. Ten (62.5%) met criteria for a modifiable mental health risk factor (i.e., low resilience, depression, or possible PTSD) at 1 month. There were

Table 5. Comparison of Patients with Mild Traumatic Brain Injury with and without Mild Post-Concussion Syndrome at 12 Months after Injury

	Mild PCS at 12 months	*No mild PCS at 12 months*	p *value*	*Cohen d*
Sample size	16	44		
Injury Severity Score: mean (SD)	4.1 (3.1)	3.9 (3.3)	0.652	0.06
Loss of Consciousness (LOC): *n* (%)	5 (31.3)	17 (38.6)	0.587	
Age: Mean (SD)	40.5 (13.4)	37.4 (12.1)	0.442	0.25
Education: mean (SD)	13.6 (3.5)	14.5 (2.9)	0.258	–0.29
Sex: female: *n* (%)	8 (50.0)	18 (40.9)	0.568	
Traumatic lesion on MRI: *n* (%)	4 (25.0)	11 (25.0)	1.00	
1 month variables				
Mild PCS+: *n* (%)	10 (62.5)	9 (20.5)	0.002	
Moderate PCS+: *n* (%)	4 (25.0)	0	0.001	
Cognitive impairment at 1 month: n (%)	6 (40.0)[1]	5 (11.6)[1]	0.025	
Fatigue (BNI-FS): mean (SD)	26.7 (20.4)	10.9 (10.9)[1]	0.003	1.13
Insomnia (ISI): mean (SD)	7.4 (6.8)	4.4 (4.2)[1]	0.178	0.60
Pain Subscale (RNBI): mean (SD)	9.8 (4.2)	7.8 (1.8)	0.093	0.76
Post-traumatic stress symptoms: mean (SD)	31.1 (12.6)	23.6 (6.5)	0.054	0.88
Depressive symptoms: mean (SD)	7.0 (6.2)	4.7 (5.4)	0.130	0.41
Quality of life (QOLIBRI): mean (SD)	140.1 (34.3)	155.8 (20.2)	0.128	–0.64
Life Satisfaction (SWLS): mean (SD)	25.4 (4.1)	26.9 (6.5)	0.183	–0.25
Resilience (RS): mean (SD)	137.7 (15.4)	141.4 (14.5)	0.328	–0.25
6 month variables				
Mild PCS+: *n* (%)	12 (75.0)	2 (4.7)	0.000	
Moderate PCS+: *n* (%)	2 (12.5)	0	0.070	
Cognitive Impairment at 6 months: *n* (%)	3 (7.5)[1]	1 (6.7)[4]	1.00	
Fatigue (BNI-FS): mean (SD)	18.3 (17.0)	5.1 (6.8)	0.002	1.26
Insomnia (ISI): mean (SD)	6.0 (4.2)	3.1 (2.8)	0.006	0.90
Pain Subscale (RNBI): mean (SD)	7.8 (3.4)	7.1 (1.7)	0.675	0.31
Post-traumatic stress symptoms: Mean (SD)	29.2 (11.5)[1]	21.5 (6.0)	0.001	1.00
Depressive symptoms: mean (SD)	6.5 (7.3)[1]	3.2 (5.3)[2]	0.007	0.56
Quality of Life (QOLIBRI): mean (SD)	147.1 (25.5)	162.5 (15.7)	0.016	–0.82
Life Satisfaction (SWLS): mean (SD)	24.7 (5.4)	26.9 (5.7)	0.092	–0.39
Resilience (RS): mean (SD)	135.6 (23.2)	143.0 (16.9)[1]	0.375	–0.39
12 month variables				
Fatigue (BNI-FS): mean (SD)	22.1 (15.9)	3.0 (5.4)	0.000	2.05
Insomnia (ISI): mean (SD)	7.8 (4.3)	3.1 (3.2)	0.000	1.34
Pain Subscale (RNBI): mean (SD)	8.6 (4.0)	6.7 (1.7)	0.341	0.76
Post-traumatic stress symptoms: mean (SD)	27.3 (8.5)	21.8 (5.4)	0.003	0.87
Quality of Life (QOLIBRI): mean (SD)	146.4 (22.6)	162.8 (19.5)	0.003	–0.81
Life Satisfaction (SWLS): mean (SD)	24.9 (4.4)	27.7 (5.5)	0.038	–0.54
Resilience (RS): mean (SD)	135.3 (17.6)	146.2 (18.0)	0.040	–0.61
Functional outcome				
Not returned to work: *n* (%)	2 (12.5)	0	0.068	
Return to work in days: mean (SD)	146.4 (295.2)	26.1 (40.0)	0.014	0.88

PCS, post-concussion syndrome; SD, standard deviation; MRI, magnetic resonance imaging; BNI-FS, Barrow Neurological Institute Fatigue Scale; ISI, Insomnia Severity Index; RNBI, Ruff Neurobehavioral Inventory; QOLIBRI, Quality of life after Brain Injury; SWLS, Satisfaction with Life Scale; RS, Resilience Scale.

[1]1 case missing, [2]4 cases missing.

The proportions were compared with Fisher exact test, and continuous variables were tested with Mann-Whitney or *t* tests.

five control subjects who met criteria for mild PCS at 12 months, and all five met criteria for possible PTSD or depression at 1 month.

Of the three patients with MTBI not returning to work within 12 months, one had trauma-related abnormalities on MRI, and two had cognitive impairment at 1 and 6 months post-injury, and two reported elevated symptoms of depression and/or PTSD at some point. One of these patients had mild PCS and one moderate PCS at 12 months; one did not answer questionnaires at 12 months.

Discussion

This study uniquely described recovery from MTBI across multiple domains in a prospective inception cohort with no pre-existing health conditions. By 6 months after injury, patients with MTBI did not differ as a group from nonhead injury trauma controls on cognition, fatigue, or mental health, and by 12 months, their level of post-concussion symptoms and quality of life was similar to that of controls. Examining individual patient outcomes, almost all patients (93.3%) with MTBI achieved favorable recovery by 1 year after injury (i.e., they did not have moderate PCS, and they had returned to work). These findings support that MTBI has a good prognosis, at least in patients who were healthy before injury.[1]

Recovery in our cohort was more rapid and complete than in some previous studies.[45–47] There are several possible reasons. First, a highly selected sample of previously healthy adults was used to minimize confounding factors. Second, cultural issues, such as symptom expectations after a head injury, have been shown to differ between countries and possibly contribute to development of persistent symptoms.[48] This study did not include an intervention, and the patients were treated according to the normal protocol after their assessments. All the patients, however, received information about their injury at the visit with the physician and neuropsychologist, and it could be hypothesized that this could have affected their symptom expectations.

Third, compensational issues can affect the results because litigation, for example, is associated with less favorable recovery from MTBI.[49] Exact data about litigation of this study population are not available. In the Finnish system, however, injured patients receive government compensation for their time off work, and during the first year after injury, personal litigation is very uncommon. Fourth, the mechanism and context of MTBI have to be considered. After car accidents, for example, bodily injuries, psychological traumatization, and access to compensation are more common than in other injury mechanisms, whereas outcome from sport-related injuries is usually better.[49] In this study there were a similar number of sport-related injuries ($n=13$, 17.6%) and car accidents ($n=12$, 16.2%).

Fifth, it has been suggested that the patients who do not attend follow-ups have a more favorable outcome than those who do.[50] Thus, it can be hypothesized that the low dropout rate in this study resulted in more recovered patients included in the 6- and 12-month follow-ups. Finally, in this study outcome was carefully examined in context (i.e., compared with that of trauma controls), thus avoiding the overestimation of the significance of mild nonspecific symptoms.

Another novel finding in the present study was that recovery was not uniform across the outcome domains. The multidimensional outcome battery and longitudinal design of this study allowed an examination of the trajectories of different domains of recovery. We found that quality of life and satisfaction with life were stable across all time points and similar between groups. Most outcome measures showed recovery and plateau by 6 months post-injury, and RTW often preceded recovery in other domains. A large proportion of patients who had returned to work within 1 month had PCS, cognitive impairment, or mental health problems at the 1-month assessment. Simply put, it was common for people to return to work while still symptomatic. It has been noted that recovery from mild head injury is better and faster when measured by functional status and quality of life than by post-concussion symptoms.[51] According to Heitger and coworkers,[51] this discrepancy between post-concussion symptoms and relatively normal functionality and quality of life could suggest that the ''good recovery'' from MTBI may involve a behavioral adaptation rather than a complete return to a previous health status.

As in many previous studies, we found that a significant minority of patients report mild post-concussion–like symptoms at 1 year (i.e., 27%). Importantly, our multidimensional assessment battery and trauma control group help provide context for this finding; 17% of those with ankle injuries also reported mild post-concussion–like symptoms at 1 year. Patients reporting ongoing mild PCS at the 12-month follow-up did not have more severe brain injuries than those without PCS (as reflected by LOC or MRI findings). They also did not have more severe bodily injuries or chronic pain. They were much slower to return to work, but virtually all were able to return within 1 year. They had higher rates of cognitive impairment at 1 month but no longer at 6 months post-injury. A large percentage (62.5%) had a modifiable psychological risk factor at 1 month (i.e., depression, possible PTSD, and/or low resilience), and at 6 months, they had greater PCS, fatigue, insomnia, traumatic stress, and depression, and worse quality of life.

In summary, patients with chronic mild PCS in our cohort achieved cognitive and functional recovery but continued to be psychologically distressed and dissatisfied. Similarly, all five of the control subjects who had mild PCS at 12 months also had a mental health problem (i.e., depression, traumatic stress, or both). Thus, it can be hypothesized that interventions targeting these variables could be beneficial for recovery. Our findings concur with previous findings that patients with persisting post-concussion symptoms have a high prevalence of comorbid psychiatric problems,[52] even though we excluded patients with pre-morbid psychiatric problems. Unlike the majority of previous studies,[53,54] for reasons that are unclear, sex was not associated with chronic PCS in this study.

Our study design does not allow us to discern the cause(s) of chronic PCS. Although chronic PCS, depression, and traumatic stress were associated, it is not clear whether emotional problems are responsible for chronic PCS, PCS results in emotional consequences, or the associations between post-concussion symptom and psychiatric symptom reporting are because of shared method variance (i.e., self-report) or etiology (i.e., brain injury). The finding that depression, traumatic stress, and PCS did not differ between patients with head versus ankle injuries at 12 months after injury makes it less likely that underlying brain injury accounts for these outcomes. As well, patients with MTBI and chronic PCS did not have more severe brain injuries, at least as indexed by MRI.

This study has limitations. First, the sample size of the study was relatively small and like in most longitudinal MTBI studies, follow-up was incomplete. Participants with missing follow-up data, however, did not differ from those who completed all assessments with respect to demographic or trauma-related (GCS, LOC) variables, symptoms at 1 month, or RTW suggesting minimal systematic bias from attrition. Second, the sample was highly selected after excluding patients with pre-existing psychiatric or neurological disorders, substance abuse, or brain injuries. Examining a previously healthy group, however, minimizes pre-injury confounding factors. We have also discussed the implications of the sample selection to drawing clinical interpretations in detail in another publication of our study group.[16]

Third, this study lacks a 3-month follow-up, which has been considered important because the symptoms lasting at least 3 months are part of the diagnostic criteria for PCS in DSM-IV.[1] Fourth, the controls only participated in neuropsychological examination once, whereas in patients with MTBI, possible practice effects can affect the results in the second examination. Also, in future studies of cognition after MTBI, it would be beneficial to have healthy controls in addition to trauma controls, for whom pain, for example, can affect symptom reporting and test performance.

The strength of this study is a homogenous study population with all the patients meeting the criteria for MTBI proposed by World Health Organization's Collaborating Centre for Neurotrauma Task Force.[15] In this study, all the participants were evaluated in the ED and underwent head CT, and 20% had trauma-related findings on brain MRI; therefore, as a group, they likely represent more seriously injured cases.[2] Thus, the favorable outcome reported was observed even when the mildest of the MTBI cases might have been left out because of not being evaluated in the ED.

The definitions of ''poor outcome'' after MTBI vary among studies, causing considerable variability in the estimates.[49] For example, some studies base poor outcome only on symptom reporting whereas others consider only persons whose social and occupational functioning is impacted to have poor outcome.[49] The effect of these different criteria still remains unclear.[49] The other strength of this study is the careful evaluation of outcome broadly, including not only symptoms, but also cognitive and mental health outcome, quality of life, and RTW during a 12-month follow-up.

Together these results support the favorable prognosis of MTBI in previously healthy adults. The results also illustrate the potential importance of providing evidence-supported treatment and rehabilitation services early in the recovery period, because those individuals who have mild persistent long-term symptoms had more severe post-concussion symptoms at 1 month, and they had modifiable psychological problems throughout the first year (e.g., traumatic stress, depression, and low resilience).

Acknowledgments

This study was done as part of the first author's PhD thesis research program. This research was funded by the Signe and Ane Gyllenberg Foundation. GLI notes that he was supported in part by the INTRuST Posttraumatic Stress Disorder and Traumatic Brain Injury Clinical Consortium funded by the Department of Defense Psychological Health/Traumatic Brain Injury Research Program (X81XWH-07-CC-CSDoD). The authors thank research assistants Anne Simi and Marika Suopanki-Ervasti for their contribution in data collection.

Author Disclosure Statement

The authors alone are responsible for the content and writing of the article. GLI has been reimbursed by the government, professional scientific bodies, and commercial organizations for discussing or presenting research relating to mild TBI and sport-related concussion at meetings, scientific conferences, and symposiums. He has a clinical and consulting practice in forensic neuropsychology involving individuals who have sustained mild TBIs. He is a co-investigator, collaborator, or consultant on grants relating to mild TBI. The other authors report no competing or conflicts of interest.

This study has been submitted to be presented at The Eleventh World Congress on Brain Injury in the Hague, March 2016.

References

1. Cassidy, J.D., Cancelliere, C., Carroll, L.J., Côté, P., Hincapié, C.A., Holm, L.W., Hartvigsen, J., Donovan, J., Nygren-de Boussard, C., Kristman, V L., and Borg, J. (2014). Systematic review of self-reported prognosis in adults after mild traumatic brain injury: results of the International Collaboration on Mild Traumatic Brain Injury Prognosis. Arch. Phys. Med. Rehabil. 95, Suppl 3, S132–S151.
2. Kristman, V.L., Borg, J., Godbolt, A.K., Salmi, L.R., Cancelliere, C., Carroll, L.J., Holm, L.W., Nygren-de Boussard, C., Hartvigsen, J., Abara, U., Donovan, J., and Cassidy, J.D. (2014). Methodological issues and research recommendations for prognosis after mild traumatic brain injury: results of the International Collaboration on Mild Traumatic Brain Injury Prognosis. Arch. Phys. Med. Rehabil. 95, Suppl 3, S266–S277.
3. Carroll, L J., Cassidy, J.D., Cancelliere, C., Côté, P., Hincapié, C.A., Kristman, V.L., Holm, L.W., Borg, J., Nygren-de Boussard, C., and Hartvigsen, J. (2014). Systematic review of the prognosis after mild traumatic brain injury in adults: cognitive, psychiatric, and mortality outcomes: results of the International Collaboration on Mild Traumatic Brain Injury Prognosis. Archives of Physical Medicine and Rehabilitation 93, Suppl 3, S152-S173.
4. Meares, S., Shores, E.A., Taylor, A.J., Batchelor, J., Bryant, R.A., Baguley, I.J., Chapman, J., Gurka, J., Capon, L., and Marosszeky, J.E. (2008). Mild traumatic brain injury does not predict acute post-concussion syndrome. J. Neurol. Neurosurg. 79, 300–306.
5. Karr, J. E., Areshenkoff, C. N., and Garcia-Barrera, M. A. (2014). The neuropsychological outcomes of concussion: a systematic review of meta-analyses on the cognitive sequelae of mild traumatic brain injury. Neuropsychology 28, 321–336.
6. Rohling, M. L., Larrabee, G. J., and Millis, S.R. (2012). The ''Miserable Minority'' following mild traumatic brain injury: who are they and do meta-analyses hide them? Clin. Neuropsychol. 26, 197–213.
7. Bigler, E.D., Farrer, T.J., Pertab, J.L., James, K., Petrie, J.A., and Hedges, D.W. (2013). Reaffirmed limitations of meta-analytic methods in the study of mild traumatic brain injury: a response to Rohling et al. Clin. Neuropsychol. 27, 176–214.
8. Cancelliere, C., Kristman, V.L., Cassidy, J.D., Hincapié, C.A., Côté, P., Boyle, E., Carroll, L.J., Stålnacke, B.M., Nygren-de Boussard, C., and Borg, J. (2014). Systematic review of return to work after mild traumatic brain injury: results of the International Collaboration on Mild Traumatic Brain Injury Prognosis. Arch. Phys. Med. Rehabil. 95, Suppl 3, S201–S209.
9. Bahraini, N.H., Breshears, R.E., Hernandez, T.D., Schneider, A.L., Forster, J.E., and Brenner, L.A. (2014). Traumatic brain injury and posttraumatic stress disorder. Psychiatr. Clin. North Am. 37, 55–75.
10. Bryant, R.A., O'Donnell, M.L., Creamer, M., McFarlane, A.C., Clark, C.R., and Silove, D. (2010). The psychiatric sequelae of traumatic injury. Am. J. Psychiatry 167, 312–320.
11. Meares, S., Shores, E.A., Taylor, A.J., Batchelor, J., Bryant, R.A., Baguley, I.J., Chapman, J., Gurka, J., and Marosszeky, J.E. (2011). The prospective course of postconcussion syndrome: the role of mild traumatic brain injury. Neuropsychology 25, 454–465.
12. Petchprapai, N., and Winkelman, C. (2007). Mild traumatic brain injury: determinants and subsequent quality of life. A review of the literature. J. Neurosci. Nurs. 39, 260–272.
13. Beseoglu, K., Roussaint, N., Steiger, H.J., and Hanggi, D. (2013). Quality of life and socio-professional reintegration after mild traumatic brain injury. Br. J. Neurosurg. 27, 202–206.
14. Ingebrigtsen, T., Romner, B., and Kock-Jensen, C. (2000). Scandinavian guidelines for initial management of minimal, mild, and moderate head injuries. The Scandinavian Neurotrauma Committee. J. Trauma 48, 760–766.
15. Holm, L., Cassidy, J.D., Carroll, L.J., Borg, J., and Neurotrauma Task Force on Mild Traumatic Brain Injury of the WHO Collaborating Centre. (2005). Summary of the WHO Collaborating Centre for Neurotrauma Task Force on Mild Traumatic Brain Injury. J. Rehabil. Med. 37, 137–141.
16. Luoto, T.M., Tenovuo, O., Kataja, A., Brander, A., Öhman, J., and Iverson, G.L. (2013). Who gets recruited in mild traumatic brain injury research? J. Neurotrauma 30, 11–16.

17. Levin, H.S., O'Donnell, V.M., and Grossman, R G. (1979). The Galveston Orientation and Amnesia Test. A practical scale to assess cognition after head injury. J. Nerv. Ment. Dis. 167, 675–684.
18. Baker, S., O'Neill, B., Haddon, W.J. Jr., and Long, W. (1974). The Injury Severity Score: a method for describing patients with multiple injuries and evaluating emergency care. J. Trauma 14, 187–196.
19. Wilson, J.T., Pettigrew, L.E., and Teasdale, G.M. (1998). Structured interviews for the Glasgow Outcome Scale and the extended Glasgow Outcome Scale: guidelines for their use. J. Neurotrauma 15, 573–585.
20. King, N.S., Crawford, S., Wenden, F.J., Moss, N.E., and Wade, D.T. (1995). The Rivermead Post Concussion Symptoms Questionnaire: a measure of symptoms commonly experienced after head injury and its reliability. J. Neurol. 242, 587–592.
21. Ashley, J. (1990). The International Classification of Diseases: the structure and content of the Tenth Revision. Health Trends 22, 135–137.
22. Wäljas, M., Iverson, G.L., Hartikainen, K.M., Liimatainen, S., Dastidar, P., Soimakallio, S., Jehkonen, M., and Öhman, J. (2012). Reliability, validity and clinical usefulness of the BNI fatigue scale in mild traumatic brain injury. Brain Inj. 26, 972–978.
23. Borgaro, S.R., Gierok, S., Caples, H., and Kwasnica, C. (2004). Fatigue after brain injury: initial reliability study of the BNI Fatigue Scale. Brain Inj. 18, 685–690.
24. Morin, C.M., Belleville, G., Bélanger, L., and Ivers, H. (2011). The Insomnia Severity Index: psychometric indicators to detect insomnia cases and evaluate treatment response. Sleep 34, 601–608.
25. Ruff, R.M., and Hibbard, K M. (2003). *Ruff Neurobehavioral Inventory Professional Manual.* Psychological Assessment Resources: Lutz, FL.
26. Weathers, F.W., Litz, B.T., Herman, D.S., Huska, J.A., and Keane, T.M. (1993). The PTSD Checklist (PCL): Reliability, validity, and diagnostic utility. Presented at the 9th Annual Conference of the ISTSS, San Antonio.
27. Ruggiero, K J., Del Ben, K., Scotti, J.R., and Rabalais, A.E. (2003). Psychometric Properties of the PTSD Checklist—Civilian Version. J. Trauma. Stress 16, 495–502.
28. American Psychological Association. (1994). *Diagnostic and Statistical Manual of Mental Disorders.* 4th ed. American Psychiatric Association: Washington DC.
29. Beck, A.T., Steer, R.A., and Brown, G.K. (1996). *Manual for the Beck Depression Inventory-II* (Finnish version). The Psychological Corporation: San Antonio, TX.
30. Wagnild, G.M., and Young, H.M. (1993). Development and psychometric evaluation of the Resilience Scale. J. Nurs. Meas. 1, 165–178.
31. Bullinger, M., Azouvi, P., Brooks, N., Basso, A., Christensen, A.L., Gobiet, W., Greenwood, R., Hutter, B., Jennett, B., Maas, A., Truelle, J.L., von Wild, K. R., and TBI Consensus Group. (2002). Quality of life in patients with traumatic brain injury—basic issues, assessment and recommendations. Restor. Neurol. Neurosci. 20, 111–124.
32. von Steinbüchel, N., Wilson, L., Gibbons, H., Hawthorne, G., Höfer, S., Schmidt, S., Bullinger, M., Maas, A., Neugebauer, E., Powell, J., von Wild, K., Zitnay, G., Bakx, W., Christensen, A.L., Koskinen, S., Sarajuuri, J., Formisano, R., Sasse, N., Truelle, J.L., and The QOLIBRI Task Force. (2010). Quality of Life after Brain Injury (QOLIBRI): scale development and metric properties. J. Neurotrauma 27, 1167–1185.
33. von Steinbüchel, N., Wilson, L., Gibbons, H., Hawthorne, G., Höfer, S., Schmidt, S., Bullinger, M., Maas, A., Neugebauer, E., Powell, J., von Wild, K., Zitnay, G., Bakx, W., Christensen, A., Koskinen, S., Formisano, R., Sarajuuri, J., Sasse, N., Truelle, J., and The QOLIBRI Task Force. (2010). Quality of Life after Brain Injury (QOLIBRI): scale validity and correlates of quality of life. J. Neurotrauma 27, 1157–1165.
34. Diener, E., Emmons, R.A., Larsen, R.J., and Griffin, S. (1985). The Satisfaction With Life Scale. J. Pers. Assess. 49, 71–75.
35. Lezak, M., Hovieson, D., and Loring, D. (2004). *Neuropsychological Assessment.* 4th ed. Oxford University Press: New York.
36. Army Individual Test Battery. (1944). *Manual of Directions and Scoring.* War Department, Adjutant General Office: Washington DC.
37. Strauss, E., Sherman, E. M., and Spreen, O. (2006). *A Compendium of Neuropsychological Tests. Administration, Norms, and Commentary.* 3rd ed. Oxford University Press: New York.
38. Halstead, W.C. (1947). *Brain and Intelligence.* University of Chicago Press: Chicago.
39. Wechsler, D. (ed) (1955). *Wechsler Adult Intelligence Scale.* (Finnish Edition: Wechslerin aikuisten älykkyysasteikko, käsikirja. Psykologien Kustannus Oy, Helsinki, 2005). 3 rd Edition. Psychological Corp: New York.
40. Mitrushina, M., Boone, K.B., Razani, J., and D'Elia, L.F. (2005). *Handbook of Normative Data for Neuropsychological Assessment*. Oxford University Press: Oxford.
41. Binder, L.M., Iverson, G.L., and Brooks, B.L. (2009). To err is human: ''abnormal'' neuropsychological scores and variability are common in healthy adults. Arch. Clin. Neuropsychol. 24, 31–46.
42. Brooks, B.L., Iverson, G.L., Feldman, H.H., and Holdnack, J.A. (2009). Minimizing misdiagnosis: psychometric criteria for possible or probable memory impairment. Dement. Geriatr. Cogn. Disord. 27, 439–450.
43. Mistridis, P., Egli, S.C., Iverson, G.L., Berres, M., Willmes, K., Welsh-Bohmer, K.A., and Monsch, A.U. (2015). Considering the base rates of low performance in cognitively healthy older adults improves the accuracy to identify neurocognitive impairment with the Consortium to Establish a Registry for Alzheimer's Disease-Neuropsychological Assessment Battery (CERAD-NAB). Eur. Arch. Psychiatry Clin. Neurosci. 265, 407–417.
44. Iverson, G L., Brooks, B.L., and Holdnack, J.A. (2012). Evidence-based neuropsychological assessment following work-related injury, in: *Neuropsychological Assessment of Work-Related Injuries.* S.S. Bush, and G L. Iverson (eds), Guilford Press: New York, pps. 360–400.
45. Cassidy, J. D., Boyle, E., and Carroll, L. J. (2014). Population-based, inception cohort study of the incidence, course, and prognosis of mild traumatic brain injury after motor vehicle collisions. Arch. Phys. Med. Rehabil. 95, Suppl 3, S278–S285.
46. Sigurdardottir, S., Andelic, N., Roe, C., Jerstad, T., and Schanke, A.K. (2009). Post-concussion symptoms after traumatic brain injury at 3 and 12 months post-injury: a prospective study. Brain Inj. 23, 489–497.
47. Thornhill, S., Teasdale, G.M., Murray, G.D., McEwen, J., Roy, C.W., and Penny, K.I. (2000). Disability in young people and adults one year after head injury: prospective cohort study. BMJ 320, 1631–1635.
48. Ferrari, R., Obelieniene, D., Russell, A. S., Darlington, P., Gervais, R., and Green, P. (2001). Symptom expectation after minor head injury. A comparative study between Canada and Lithuania. Clin. Neurol. Neurosurg. 103, 184–190.
49. Iverson, G.L., Silverberg, N., Lange, R.T., and Zasler, N.D. (2012). Conceptualizing outcome from mild traumatic brain injury, in: *Brain Injury Medicine: Principles and Practice,* 2nd ed. N.N. Zasler, D.I. Katz, and R.D. Zafonte (eds). Demos Medical Publishing: New York, pps. 470–497.
50. Vikane, E., Hellstrom, T., Roe, C., Bautz-Holter, E., Assmus, J., and Skouen, J. S. (2014). Missing a follow-up after mild traumatic brain injury—does it matter? Brain Inj. 28, 1374–1380.
51. Heitger, M.H., Jones, R.D., Frampton, C.M., Ardagh, M.W., and Anderson, T.J. (2007). Recovery in the first year after mild head injury: divergence of symptom status and self-perceived quality of life. J. Rehabil. Med. 39, 612–621.
52. King, N.S., and Kirwilliam, S. (2011). Permanent post-concussion symptoms after mild head injury. Brain Inj. 25, 462–470.
53. King, N.S. (2014). A systematic review of age and gender factors in prolonged post-concussion symptoms after mild head injury. Brain Inj. 28, 1639–1645.
54. Silverberg, N.D., Gardner, A.J., Brubacher, J. R., Panenka, W J., Li, J.J., and Iverson, G.L. (2015). Systematic review of multivariable prognostic models for mild traumatic brain injury. J. Neurotrauma 32, 517–526.

Address correspondence to:
Heidi Losoi, Psych L
Department of Neurosciences and Rehabilitation
Tampere University Hospital
PO Box 2000
Tampere FI-33521
Finland

E-mail: heidi.losoi@gmail.com